

PRODUCT REQUIREMENTS DOCUMENT (PRD)

Project Title

KAVYA-KANAJA

AI-Enhanced Kannada Poetry Learning Android Application

Prepared By

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1. DOCUMENT INFORMATION

Field	Details
Document Title	Product Requirements Document (PRD)
Product Name	Kavya-Kanaja
Version	1.0
Date	May 2026
Status	Approved for Development
Author	Adarsha R (1SJ22EC004)
Internal Guide	Dr. C. Rangaswamy, HOD & Professor, Dept. of ECE, SJCIT
External Mentor	Tirumal Mutalikdesai, Mentor, MindMatrix.io, Bangalore
College	S.J.C. Institute of Technology, Chikkaballapur – 562103
University	Visvesvaraya Technological University (VTU), Belagavi
Internship Org	MindMatrix.io, Bangalore, Karnataka
Platform	Android (API Level 21 – 34)
Primary Language	Kotlin + Jetpack Compose
Domain	Education — Kannada Poetry Learning & Literary Appreciation

2. EXECUTIVE SUMMARY

Kavya-Kanaja (meaning 'Treasury of Poetry' in Kannada) is an AI-enhanced Android application designed to make Kannada classical poetry accessible, meaningful, and engaging for learners of all ages. The application bridges a critical cultural and educational gap in Karnataka's digital landscape, where Kannada literature — particularly the rich tradition of classical poetry — remains largely inaccessible in interactive, bilingual digital formats.

The application provides a curated collection of 25+ poems by celebrated Jnanpith Award-winning Kannada poets such as Kuvempu, D.R. Bendre, and V.K. Gokak, categorised across themes of nature, devotion, love, patriotism, philosophy, and social commentary. It integrates Google Gemini AI for deep literary explanations, Android TextToSpeech for bilingual audio recitation with real-time word highlighting, a Kannada-English dictionary, Bhavartha (meaning and interpretation) screens, and Firebase-powered Google Sign-In authentication.

This PRD defines all product requirements, user personas, features, user stories, acceptance criteria, and success metrics that guide the development of Version 1.0 of Kavya-Kanaja.

3. PROBLEM STATEMENT

3.1 The Problem

Karnataka's rich tradition of classical Kannada poetry — spanning centuries of Jnanpith Award-winning literature — remains largely inaccessible to younger generations, students, and non-native learners. Existing digital resources are English-centric, lack interactive exploration, provide no audio guidance, and offer no AI-powered literary interpretation. Students struggle with the complex vocabulary and metaphorical depth of classical Kannada verse, while the cultural significance of these works is lost without proper context and guided understanding.

Existing apps fail these users because they require constant internet access for basic functions, offer no Kannada audio recitation, provide no AI-driven literary explanation, and present information in static, non-interactive formats that are disengaging for modern learners.

3.2 Problem Validation

Problem Area	Evidence / Impact
No Dedicated Kannada Poetry App	Zero apps exist for Kannada classical poetry with AI explanations and bilingual audio recitation.
Language Accessibility	Karnataka has 60M+ Kannada speakers. Students need bilingual support to understand classical verse vocabulary.
No AI Literary Guidance	Poems by Kuvempu, Bendre and others contain deep philosophical references inaccessible without expert guidance.
No Audio Learning	Students cannot appreciate the musicality of Kannada verse without hearing correct pronunciation and rhythm.
Poor Poet Discovery	Biographical and literary significance of Jnanpith laureates not available in any single curated mobile app.
Static Content Only	All existing resources offer plain text with no interactivity — no word-tap meaning, no AI explanation, no audio.

3.3 Goal Statement

Build an AI-enhanced, bilingual Android application that makes Kannada classical poetry accessible, comprehensible, and culturally enriching for every learner — through curated poem collections, real-time audio recitation, AI-powered literary analysis, and an interactive Kannada-English dictionary — precisely when and where curiosity meets language.

4. USER PERSONAS

4.1 Primary Persona — Kannada Literature Student

Attribute	Description
Name	Kavitha (Illustrative)
Age	19
Location	Chikkaballapur district, Karnataka
Language	Kannada (primary), basic English
Device	Mid-range Android smartphone (4 GB RAM, Android 10)
Connectivity	Moderate 4G connectivity
Scenario	Studying classical Kannada poetry for Pre-University examinations
Pain Points	Dense classical vocabulary, no bilingual explanations, no audio for pronunciation
Goal	Understand poem meaning, Bhavartha, and literary significance with AI assistance

4.2 Secondary Persona — Kannada Language Enthusiast

Attribute	Description
Name	Ravi (Illustrative)
Age	32
Location	Bengaluru, Karnataka

Attribute	Description
Language	Kannada and English
Device	High-range Android smartphone
Connectivity	Reliable 4G/WiFi
Scenario	Cultural enthusiast wanting to reconnect with Kannada literary heritage
Pain Points	No single curated platform for classical poems with poet bios and AI explanation
Goal	Explore Kannada poetry daily, discover new poems, deepen literary appreciation

4.3 Tertiary Persona — Kannada Teacher

Attribute	Description
Name	Suma (Illustrative)
Age	42
Location	Tumkur district, Karnataka
Language	Kannada (primary), English
Device	Mid-range Android smartphone (4 GB RAM, Android 11)
Connectivity	Moderate WiFi in school, 4G in field
Scenario	Uses the app as a classroom reference for teaching Kannada poetry
Pain Points	No digital teaching aid with bilingual audio and AI literary analysis
Goal	Access curated poems with Bhavartha and AI explanations to enrich teaching

5. PRODUCT SCOPE

5.1 In Scope — Version 1.0

- Curated collection of 25+ Kannada poems by 10 Jnanpith Award-winning poets.
- Daily poem feature — a new poem surfaced each day via date-indexed selection.
- Bilingual interface supporting Kannada and English across all screens.
- Android TextToSpeech-powered audio recitation with real-time word highlighting.
- Adjustable speech speed — Slow (0.6x), Normal (1.0x), Fast (1.4x) — for personalised learning.
- Google Gemini AI-powered literary explanation (theme, literary devices, significance, life lesson).
- Bhavartha screen — original Kannada text, English meaning, and Bhavartha interpretation.
- Kannada-English word dictionary with contextual AI word meaning via Gemini.
- Poet biography screen with life history, awards, birthplace, and famous works.
- Category-based poem browsing — Nature, Devotion, Love, Patriotism, Philosophy, Social, Mother.
- Firebase Google Sign-In authentication with animated login screen.
- MVVM architecture with StateFlow and Repository pattern.
- Dark/Light theme support via KavyaTheme system.

5.2 Out of Scope — Version 1.0

- User-generated poem submissions or community content.
- Offline AI using on-device Gemini Nano (requires Android AICore — future version).
- Multi-language support beyond Kannada and English.
- Quiz or gamification features for poetry learning.

- Social sharing of poems or user cloud sync.
- AR-guided poetry recitation or visual overlays.
- Wearable / Wear OS companion application.
- Push notifications for daily poem reminder.

6. FEATURES & USER STORIES

6.1 Feature F-01: Daily Poem & Poem Collection

Core content feature. Delivers a curated daily poem from a collection of 25+ Kannada poems by Jnanpith Award-winning poets, with category browsing and date-indexed daily poem selection.

ID	User Story	Acceptance Criteria
US-01	As a user, I want to see today's poem on the home screen so I discover a new poem each day.	Home screen displays daily poem with title, poet name, and first verse. Poem changes each day via date index.
US-02	As a user, I want to browse poems by category so I can find poems matching my interest.	Category filter displays poems across: Nature, Devotion, Love, Patriotism, Philosophy, Social, Mother.
US-03	As a user, I want to see the poet's name alongside the poem.	Poet name displayed on poem card. Tapping navigates to poet biography screen.
US-04	As a user, I want to read the full poem with Kannada text and English meaning.	Full poem screen shows Kannada verse, English meaning, and Bhavartha in clearly separated cards.

6.2 Feature F-02: Audio Recitation — Listen Screen

Enables users to listen to poems recited via Android TextToSpeech in Kannada and English, with real-time word highlighting that tracks the currently spoken word.

ID	User Story	Acceptance Criteria
US-05	As a user, I want to listen to the poem in Kannada so I hear correct pronunciation.	Play button activates TTS with Locale('kn','IN'). Currently spoken word highlighted in orange with underline.
US-06	As a user, I want to listen to the English meaning so I understand the poem.	Separate Play button plays English meaning using Locale.ENGLISH. Stop halts playback.
US-07	As a user, I want to control speech speed so I can learn at my own pace.	Slow (0.6x), Normal (1.0x), Fast (1.4x) speed buttons correctly set TTS speech rate.
US-08	As a user, I want audio to stop when I leave the screen.	TTS stops on lifecycle ON_PAUSE. TTS is shut down on screen dispose.

6.3 Feature F-03: Bhavartha — Meaning & Explanation Screen

Dedicated screen for deep literary engagement. Displays original Kannada text, English meaning, and Bhavartha, with AI-powered explanation via Google Gemini.

ID	User Story	Acceptance Criteria
US-09	As a user, I want to read the Bhavartha of the poem so I understand its deeper meaning.	Screen displays Kannada text, English meaning, and Bhavartha in separate labelled cards.
US-10	As a user, I want AI to explain the poem in simple English.	'Ask AI to Explain' button calls Gemini. Response covers theme, literary devices, significance, life lesson.
US-11	As a user, I want a loading indicator while AI explanation is generated.	CircularProgressIndicator shown during API call. Result replaces loader on completion.

ID	User Story	Acceptance Criteria
US-12	As a user, I want the AI explanation in plain, simple English.	Gemini prompt instructs the model to use simple language accessible to all audiences.

6.4 Feature F-04: Kannada-English Dictionary

Offline Kannada-English word dictionary bundled within the APK, augmented by AI-powered contextual word meaning via Google Gemini.

ID	User Story	Acceptance Criteria
US-13	As a user, I want to look up Kannada words to understand difficult vocabulary.	Dictionary loads from APK-bundled JSON. Search returns Kannada word with English meaning.
US-14	As a user, I want AI to give contextual meaning of a word within the poem.	Gemini called with word and poem context. Returns 1-line contextual meaning.
US-15	As a user, I want the base dictionary to work without internet.	Base dictionary from assets/dictionary.json. Offline lookup works without network calls.

6.5 Feature F-05: Poet Biography

Detailed biographical profiles of 10 Jnanpith Award-winning Kannada poets.

ID	User Story	Acceptance Criteria
US-16	As a user, I want to read about the poet so I understand their life and contribution.	Poet profile shows: name (Kannada & English), bio, birthplace, famous works, awards, and dates.
US-17	As a user, I want to see all poems by a specific poet.	Poet profile shows associated poems. Tapping a poem navigates to its full poem screen.

6.6 Feature F-06: Authentication

ID	User Story	Acceptance Criteria
US-18	As a user, I want to sign in with my Google account so my session is secure.	Google Sign-In via Firebase authenticates user. On success, navigates to Home Screen.
US-19	As a user, I want the login screen to load smoothly with an animation.	Login screen fades in using animateFloatAsState with 1000ms tween. Smooth UX on load.

7. FUNCTIONAL REQUIREMENTS

ID	Requirement	Priority	Feature
FR-01	App shall display Home Screen with daily poem within 2 seconds of launch.	Critical	F-01
FR-02	App shall load poem collection from APK-bundled JSON assets.	Critical	F-01
FR-03	App shall support category-based poem filtering across all 7 categories.	High	F-01
FR-04	App shall display daily poem using date-index-based selection.	Critical	F-01
FR-05	App shall activate Android TTS for Kannada audio using Locale('kn','IN').	High	F-02
FR-06	App shall highlight the currently spoken word in real-time during Kannada TTS.	High	F-02
FR-07	App shall support speech speed control: Slow (0.6x), Normal (1.0x), Fast (1.4x).	High	F-02

ID	Requirement	Priority	Feature
FR-08	App shall stop TTS on screen lifecycle ON_PAUSE event.	High	F-02
FR-09	App shall display Bhavartha with Kannada text, English meaning, and interpretation.	Critical	F-03
FR-10	App shall call Gemini AI API to generate literary explanation on user request.	High	F-03
FR-11	App shall load Kannada-English dictionary from APK-bundled dictionary.json.	High	F-04
FR-12	App shall call Gemini API for contextual word meaning in poem context.	Medium	F-04
FR-13	App shall display poet biography with full profile, awards, and famous works.	High	F-05
FR-14	App shall implement Firebase Google Sign-In with animated login screen.	Critical	F-06
FR-15	App shall navigate between all screens using Jetpack Navigation Component.	High	All
FR-16	App shall support Dark and Light theme via KavyaTheme system.	Medium	All
FR-17	Gemini API key shall be managed via BuildConfig — not in source code.	Critical	F-03/04

8. NON-FUNCTIONAL REQUIREMENTS

ID	Category	Requirement
NFR-01	Performance	Cold start to Home Screen within 2 seconds on mid-range hardware.
NFR-02	Performance	Individual screen transitions within 500ms.
NFR-03	Performance	Poem and dictionary JSON fully pre-loaded and cached in-memory after first access.
NFR-04	Reliability	Zero crashes on all standard user flows.
NFR-05	Reliability	TTS gracefully handles Kannada locale unavailability with English fallback.
NFR-06	Accessibility	All interactive elements include contentDescription for TalkBack screen reader.
NFR-07	Accessibility	All UI touch targets meet 48dp minimum per Material Design 3 guidelines.
NFR-08	Compatibility	App supports Android API Level 21 (Android 5.0) through API Level 34 (Android 14).
NFR-09	Compatibility	App runs correctly on 2 GB RAM devices with no significant performance degradation.
NFR-10	Security	Gemini API key managed via BuildConfig — not committed to GitHub source code.
NFR-11	Security	Firebase Google Sign-In handles authentication securely. No credentials stored locally.
NFR-12	Maintainability	MVVM architecture with Repository abstraction — UI does not directly reference data sources.
NFR-13	Maintainability	Each feature implemented as independent Compose screen backed by dedicated ViewModel.
NFR-14	Usability	All critical actions achievable in 3 taps or fewer.

ID	Category	Requirement
NFR-15	Usability	Dark/Light theme support for comfortable reading in all lighting conditions.

9. TECHNOLOGY STACK

Technology	Purpose
Kotlin 1.9+	Primary programming language for all Android development.
Jetpack Compose BOM 2024.02+	Declarative UI toolkit for all application screens and components.
Material Design 3 (Latest stable)	Google's design system — UI components, theming, dark/light mode.
MVVM + StateFlow (Jetpack Arch)	Architecture pattern — clean separation of UI, business logic, and data.
Android TextToSpeech (Native)	On-device TTS for Kannada and English poem recitation with word highlighting.
Google Generative AI SDK (gemini-flash)	Kotlin SDK for Gemini AI — literary explanation and word meaning features.
Firebase Auth (Latest stable)	Google Sign-In authentication and user session management.
Google Sign-In SDK (Latest stable)	OAuth 2.0-based Google account authentication integration.
Gson (Latest stable)	JSON parsing for APK-bundled poem collection and dictionary data.
Jetpack Navigation (Latest stable)	Screen transitions, back stack management, argument passing.
SharedPreferences (Android native)	Persistent storage for theme preference and user settings.
Android Studio Hedgehog 2023.1.1+	Primary IDE — development, profiling, and testing.
Git / GitHub (Latest)	Version control and project repository hosting.
Google AI Studio (Free tier)	Gemini API key management for AI explanation and dictionary features.

10. UX & DESIGN REQUIREMENTS

10.1 Design Principles

- **Literary Aesthetic:** Serif typography, warm tones, and paper-like card surfaces to evoke reading a poetry book.
- **Accessibility:** 48dp minimum touch targets. High-contrast visuals. TalkBack screen reader support throughout.
- **Bilingual by Default:** Kannada script and English are equal first-class citizens on all screens.
- **AI-Assisted Discovery:** AI explanation is a tap away — never hidden behind complex menus.
- **Minimal Cognitive Load:** One poem per screen. Progressive disclosure of meaning, Bhavartha, and AI insight.
- **Dark/Light Theme:** Full theme support for comfortable reading via KavyaTheme system.

10.2 Screen Inventory

Screen	Description	Key Elements
Login Screen	Animated entry with Google Sign-In.	App logo, animated fade-in, Google Sign-In button, loading state, error handling.

Screen	Description	Key Elements
Home Screen	Daily poem card and navigation.	Today's poem card, poet name, category badges, bottom nav to Listen, Bhavartha, Dictionary.
Listen Screen	Audio recitation with word highlighting.	Kannada poem card with word highlight, English meaning card, Play/Stop, speed controls.
Bhavartha Screen	Meaning and AI explanation.	Poem title, Kannada text card, English meaning card, Bhavartha card, Ask AI button, AI card.
Dictionary	Kannada-English word lookup.	Search bar, word list from JSON, contextual AI meaning on tap.
Poet Profile	Poet biography and works listing.	Poet name (Kannada & English), bio, birthplace, awards, famous works, poems list.
Category Screen	Browse poems by theme category.	Category chips (Nature, Devotion, Love, Patriotism, Philosophy, Social, Mother), poem cards.

11. SUCCESS METRICS & KPIs

Metric	Target	Measurement Method
App cold start to Home Screen	Under 2 seconds	Android Profiler — measured on mid-range device.
Screen-to-screen navigation time	Under 500ms	Android Profiler — all screen transitions.
TTS word highlight accuracy	95%+ words highlighted	Manual test with 5 poems across both languages.
AI explanation response time	Under 5 seconds	Measured from button tap to full response render.
Dictionary lookup speed	Instant (<100ms)	JSON loaded in-memory — no file I/O on lookup.
Google Sign-In success rate	100% on valid accounts	Manual testing with Firebase test suite.
Kannada content accuracy	100% validated	Native Kannada speaker review of all poem texts and Bhavartha.
TalkBack accessibility compliance	All screens pass	TalkBack audit — all interactive elements labelled.
Theme switch performance	Instant — no restart	Theme toggle tested across all screens.
Poem collection coverage	25+ poems, 10 poets	Data verified in poems.json asset file.

12. ASSUMPTIONS & CONSTRAINTS

12.1 Assumptions

- Target users own an Android smartphone running Android 5.0 (API 21) or above with at least 2 GB RAM.
- All Kannada poem texts and Bhavartha content are validated by a native Kannada speaker before inclusion.
- Google Gemini Flash free-tier API quota is sufficient for expected usage in Version 1.0.
- Firebase Authentication free tier is adequate for Version 1.0 user volume.
- Poem content is curated from works in the public domain or licensed for educational use.
- Users are students, enthusiasts, or teachers seeking literary engagement.

12.2 Constraints

- The Gemini API key must not be committed to the GitHub repository — managed via BuildConfig.
- All core poem reading features must function with minimal internet connectivity.
- The application must support a minimum of Android API Level 21.
- Development timeline is constrained to the MindMatrix VTU internship programme duration.
- All UI must use Material Design 3 components — no custom component frameworks.
- Firebase Google Sign-In requires internet connectivity for initial authentication.

13. FUTURE ROADMAP

Phase	Feature	Version
Phase 2	Quiz & Gamification — comprehension quizzes, streak rewards, and badges.	v2.0
Phase 2	Expanded Collection — 100+ poems across all major Kannada literary eras.	v2.0
Phase 2	Push Notifications — daily poem reminder with a verse snippet.	v2.0
Phase 2	Multi-Language Support — Hindi and Telugu alongside Kannada and English.	v2.0
Phase 3	On-Device AI — Gemini Nano via Android AICore for full AI without internet.	v3.0
Phase 3	Social Features — share poem of the day, favourites, and AI explanations.	v3.0
Phase 3	Wear OS Companion App — poem of the day and word meaning on smartwatch.	v3.0
Phase 4	Community Contributions — verified Kannada scholars submit poems via moderated pipeline.	v4.0
Phase 4	AI Personalization — on-device ML to surface poems based on user reading history.	v4.0

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15. APPROVAL

Role	Name
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